

Multiple Choice

1. **Answer:** C

Options: A) They are generated when memory cycles are "stolen".; B) They are used in place of data channels.; C) They can indicate completion of an I/O operation.; D) They cannot be generated by arithmetic operations.

Note: Correct option: C - They can indicate completion of an I/O operation.

2. **Answer:** C

Options: A) Fast hardware exists to move blocks of pixels efficiently.; B) Realistic lighting and shading can be done.; C) All line segments can be displayed as straight.; D) Polygons can be filled with solid colors and textures.

Note: Correct option: C - All line segments can be displayed as straight.

3. **Answer:** A

Options: A) `int(x [,base])`; B) `long(x [,base] )`; C) `float(x)`; D) `str(x)`

Note: Correct option: A - `int(x [,base])`

4. **Answer:** B

Options: A) Finding a longest simple cycle in G; B) Finding a shortest cycle in G; C) Finding ALL spanning trees of G; D) Finding a largest clique in G

Note: Correct option: B - Finding a shortest cycle in G

5. **Answer:** C

Options: A) Insertion sort; B) Quicksort; C) Merge sort; D) Selection sort

Note: Correct option: C - Merge sort

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'one in which each main memory word can be stored at any of 3 cache locations'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'accepted by some Turing machine, but by no pushdown automaton'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Operator precedence'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Both of the above.'

Long Form Response

11. **Answer:** `def count__no (A,N,L,R): | return (i)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def remove_parenthesis(items): | return (re.sub(r" ?[)]+", "", item))`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key